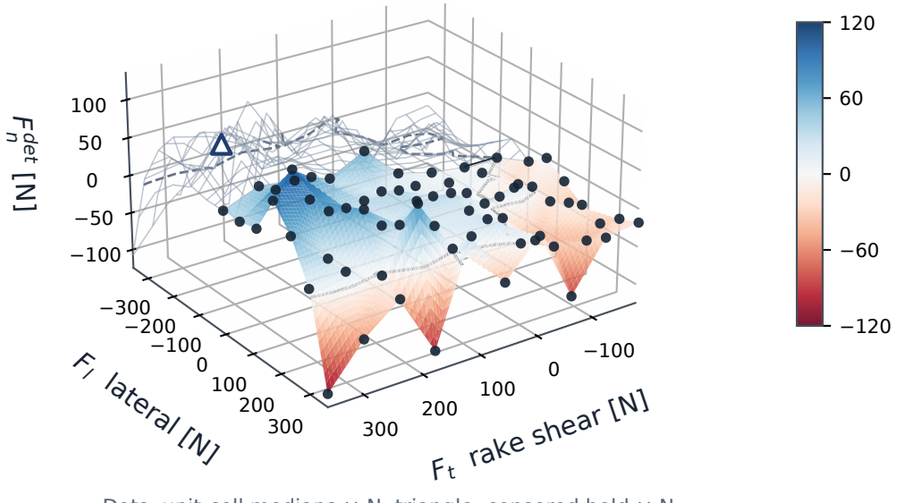
Predicted pad limit surface from unit bench tests

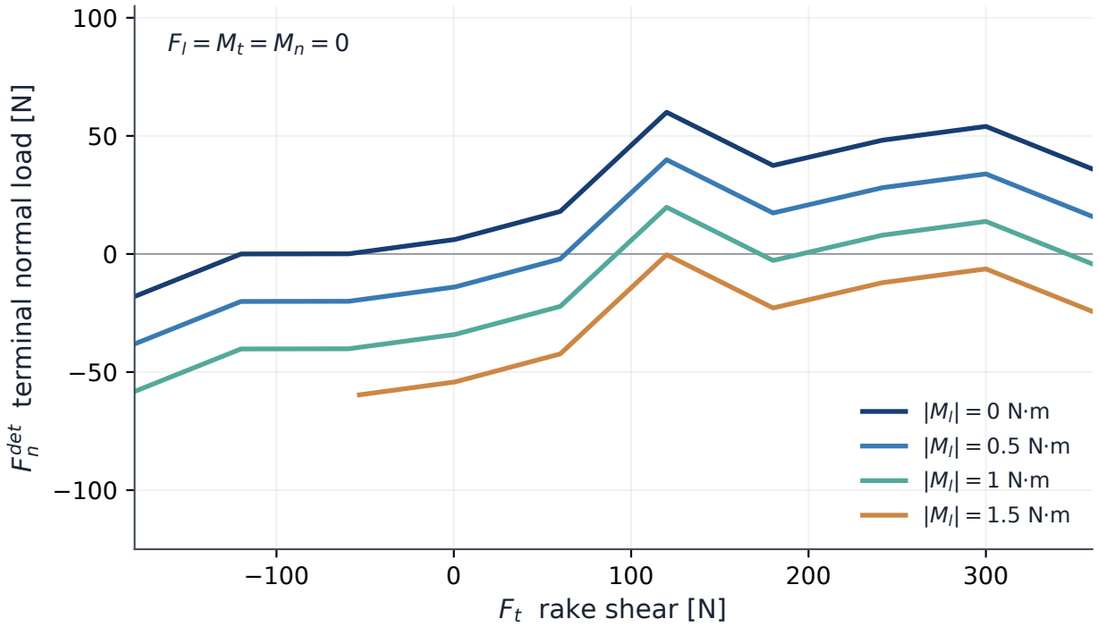
12-unit pad | 3 × 4 grid | 74.6 × 96 mm grid span
Terminal-detachment command · 0.5 N unit steps · 3 s dwell · 10 N unit preload

(a) Zero-moment force boundary

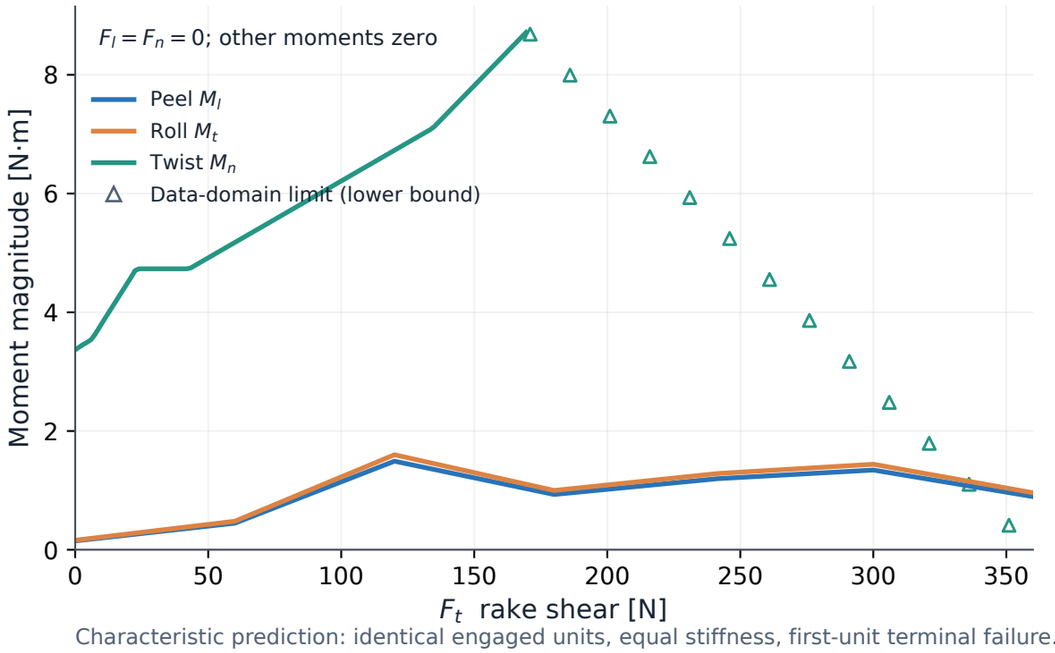


Dots: unit-cell medians × N; triangle: censored hold × N
Gray wireframe: untested lateral half, assumed symmetric

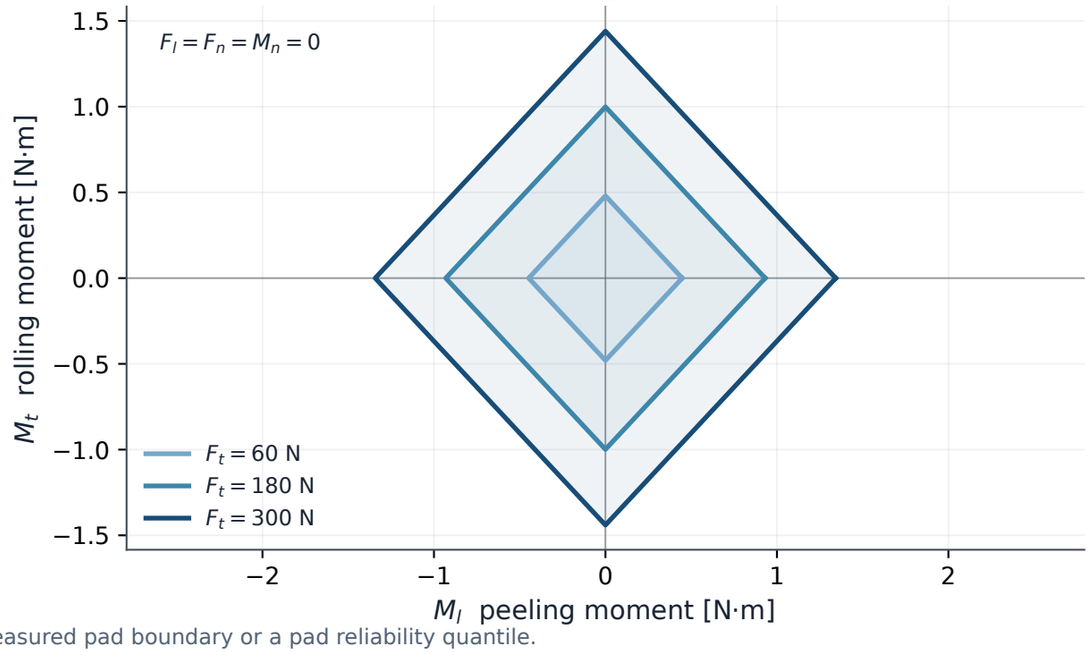
(b) Peeling reduces normal-load capacity



(c) Moment limits at zero normal load



(d) Coupled peel–roll boundary



Characteristic prediction: identical engaged units, equal stiffness, first-unit terminal failure. This is not a measured pad boundary or a pad reliability quantile.
Interpolation stays inside the tested unit-force domain; the plotted force range is not an ultimate shear limit.